

TARGET FOILS INFORMATION

ST1

Supermendur deposited on Cu foil

Weight: 0.15176g
Length: 107.8mm (averaged)
Width: 11mm
Thickness: 15.76 μ m(calculated)
Polarization: 7.961% (530Gauss)
annealed at KIPT 850°C :3+4hours, cooled to 300°C with 50°C gradient

ST2

Supermendur deposited on Cu foil

Weight: 0.24616g
Length: 107.8mm (averaged)
Width: 11mm
Thickness: 25.56 μ m(calculated)
Polarization: 7.972% (530Gauss)
annealed at KIPT 850°C :3+4hours, cooled to 300°C with 50°C gradient
saturation curve was measured

ST3

Supermendur deposited on Cu foil

Weight: 0.04644g
Length: 107.8mm (averaged)
Width: 11mm
Thickness: 4.82 μ m(calculated)
Polarization: 7.593% (530Gauss)
annealed at KIPT 850°C :3+4hours, cooled to 300°C with 50°C gradient

ST4

Supermendur deposited on Cu foil

Weight: 0.10244g
Length: 107.8mm (averaged)
Width: 11mm
Thickness: 10.64 μ m(calculated)
Polarization: 7.963% (530Gauss)
annealed at KIPT 850°C :3+4hours, cooled to 300°C with 50°C gradient

thickness.kumac, thickness.ps: ST1–ST4 foil thickness dependence

S1

Supermendur deposited no Cu

Weight: 0.0442g
Length: 9.9mm
Width: 10mm
Thickness: 5.45 μ m(calculated)
Polarization: 8.246% (530Gauss)
annealed at KIPT 850°C :3+4hours, cooled to 300°C with 50°C gradient

S2

Supermendur deposited no Cu

Weight: 0.0413g
Length: 9.6mm
Width: 10mm
Thickness: 5.3 μ m(calculated)
Polarization: 8.164% (530Gauss)
annealed at KIPT 850°C :3+4hours, cooled to 300°C with 50°C gradient
saturation curve was measured

S1S2

Supermendur deposited no Cu
superposition S1+S2

Weight: 0.0855g
Length: 9.75mm
Width: 10mm
Thickness: 5.37 μ m(calculated)
Polarization: 8.199% (530Gauss)
annealed at KIPT 850°C :3+4hours, cooled to 300°C with 50°C gradient

S1pS2

Supermendur deposited no Cu
superposition S1+isolator+S2

Weight: 0.0855g
Length: 9.75mm
Width: 10mm
Thickness: 5.37 μ m(calculated)
Polarization: 8.195% (530Gauss)
annealed at KIPT 850°C :3+4hours, cooled to 300°C with 50°C gradient

SLAC201

American Supermendur rolled (from A. Deur)

Weight: 1.7196g
Length: 364mm
Width: 30.02mm
Thickness: 19.38 μ m(calculated)
Polarization: 8.239% (530Gauss)
Polarization: 8.191% (94Gauss)
saturation curve was measured
foil was shipped to SLAC

SLAC401

American Supermendur rolled (from A. Deur)

Weight: 3.516g
Length: 364mm
Width: 30.2mm
Thickness: 39.39 μ m(calculated)
Polarization: 8.216% (530Gauss)
Polarization: 8.147% (94Gauss)
foil was used for an angular dependence measurement

S701

Supermendur **VACOFLUX48** from Vacuumshmetce
rolled and annealed at NSC KIPT

Weight: 8.786g
Length: 359.5mm
Width: 30.12mm
30.31mm
30.44mm
average: 30.28mm
Thickness: 99.35 μ m(calculated)
Polarization: 8.334% (530Gauss)
annealed at KIPT 880°C :3+4hours, cooled to 400°C with 50°C gradient
***There is a contamination on the foil surface (probably, oil from the vacuum pump in process of annealing). Right edge of the foil surface was cleaned with a thin sandpaper.
Saturation curve was measured

SC701(treated S701)

Supermendur **VACOFLUX48** from Vacuumshmetce
rolled and annealed
at NSC KIPT

Weight: 8.768g
Length: 359.5mm
Width: 30.12mm
30.31mm
30.44mm
average: 30.28mm
Thickness: 99.21μm(calculated)
Polarization: 8.297% (530Gauss)
annealed at KIPT 880°C :3+4hours, cooled to 400°C with 50°C gradient
***There was a contamination on the foil surface (probably, oil from the vacuum pump in process of annealing). Whole foil surface was cleaned with a thin sandpaper.
Saturation curve was measured.

SCC701(treated S701)

Supermendur **VACOFLUX48** from Vacuumshmetce
rolled and annealed at NSC KIPT

Weight: 8.768g
Length: 359.5mm
Width: 30.12mm
30.31mm
30.44mm
average: 30.28mm
Thickness: 99.21μm(calculated)
Polarization: 8.296% (530Gauss)
annealed at KIPT 880°C :3+4hours, cooled to 400°C with 50°C gradient
***There was a contamination on the foil surface (probably, oil from the vacuum pump in process of annealing). Whole foil surface was cleaned with a thin sandpaper. Central part of the foil was additionally treated with sandpaper
s701_sum.kumac, s701_sum.ps – summary for s701, sc701 and scc701

S98

Supermendur **VACOFLUX48** from Vacuumshmetce
rolled and annealed at NSC KIPT

Weight: 8.6139g
Length: 360.15mm
Width: 29.794mm
Thickness: 98.266μm(calculated)
Polarization: 8.339% (530Gauss)
annealed at KIPT 880°C :3+4hours, cooled to 400°C with 50°C gradient
***There is a contamination on the foil surface (probably, oil from the vacuum pump in process of annealing).
Saturation curve was measured

S96

Supremendur **VACOFLUX48** from Vacuumshmetce
rolled and annealed at NSC KIPT

Weight: 8.5638g
Length: 359.38mm
Width: 30.333mm
Thickness: 96.748 μ m(calculated)
Polarization: 8.343% (530Gauss)
annealed at KIPT 880°C :3+4hours, cooled to 400°C with 50°C gradient
***There is a contamination on the foil surface (probably, oil from the vacuum pump in process of annealing).
Saturation curve was measured.
Foil was shipped to SLAC

S96n

Supremendur **VACOFLUX48** from Vacuumshmetce
rolled and annealed at NSC KIPT
narrow foil

Weight: 3.1639g
Length: 367.36mm
Width: 11.01mm
Thickness: 96.321 μ m(calculated)
Polarization: 8.342% (530Gauss)
annealed at KIPT 880°C :3+4hours, cooled to 400°C with 50°C gradient
saturation curve was measured
***There is a contamination on the foil surface (probably, oil from the vacuum pump in process of annealing).

S189

Supremendur **VACOFLUX48** from Vacuumshmetce
annealed at NSC KIPT

Weight: 1.592g
Length: 185.9mm
Width: 5.352mm
Thickness: 197.06 μ m(calculated)
Polarization: 8.322% (530Gauss)
annealed at KIPT 880°C :3+4hours, cooled to 400°C with 50°C gradient
saturation curve was measured
***There is a contamination on the foil surface (probably, oil from the vacuum pump in process of annealing).

FH51(LFH51)

Pure Fe

deposited on Cu+Ta and annealed at NSC KIPT

Weight of Cu: 3.02825g
Weight of Cu+Ta: 3.03445g
Weight of Cu+Ta+4Fe samples: 3.27680g
Weight of Fe: 0.06059g
Length: 110mm
Width: 10mm
Thickness of Fe: 9.6 μ m(calculated)
Thickness of Ta: 850Angstrom(calculated)
Thickness of Cu: ~30 μ m
Polarization: 7.046% (530Gauss)
annealed at KIPT 800°C :1hour, cooled to 500°C with 50°C gradient
***inclination of the polarization curve may be explained with 45deg inclination of Cu foil in accordance to a cathode in process of deposition.

FH32(LFH32)

Pure Fe

deposited on Cu+Ta and annealed at NSC KIPT

Weight of Cu: 3.07165g
Weight of Cu+Ta: 3.07425g
Weight of Cu+Ta+4Fe samples: 3.40545g
Weight of Fe: 0.08280g
Length: 110mm
Width: 10mm
Thickness of Fe: 7.0 μ m(calculated)
Thickness of Ta: 360Angstrom(calculated)
Thickness of Cu: ~30 μ m
Polarization: 6.946% (530Gauss)
annealed at KIPT 710°C :1hour, cooled to 500°C with 50°C gradient

SH6SH1

Supermendur

deposited on Cu+Ta and annealed at NSC KIPT

Weight of Cu: 3.02640g
Weight of Cu+Ta: 3.03315g
Weight of Cu+Ta+4Fe samples: 3.30350g
Weight of FeCoV: 0.06761g
Length: 110mm
Width: 10mm
Thickness of FeCoV: 8.18 μ m(calculated)
Thickness of Ta: 920Angstrom(calculated)

Thickness of Cu: ~30 μ m
Polarization: 6.946% (530Gauss)
annealed at KIPT 850°C :1hour, cooled to 500°C with 50°C gradient

SH6SH2

Supermendur

deposited on Cu+Ta and annealed at NSC KIPT

Weight of Cu: 3.02640g
Weight of Cu+Ta: 3.03315g
Weight of Cu+Ta+4Fe samples: 3.30350g
Weight of FeCoV: 0.06761g
Length: 10.5mm
Width: 10mm
Thickness of FeCoV: 8.18 μ m(calculated)
Thickness of Ta: 920Angstrom(calculated)
Thickness of Cu: ~30 μ m
Polarization: 2.626% (530Gauss)
annealed at KIPT 850°C :1hour, cooled to 500°C with 50°C gradient

SA14

Supermendur

rolled and annealed at NSC KIPT

Weight: 0.507g
Length: 215.5mm
Width: 30mm
Thickness: 9.66 μ m(calculated)
Polarization: 8.026% (530Gauss)
SA14 was used for PIGE thickness measurements
saturation curve was measured

SUPA12m

American Supermendur

rolled and annealed at NSC KIPT

Weight of: 0.512g
Length: 300.0mm
Width: 30mm
Thickness: 7.01 μ m(calculated)
Polarization: 8.016% (530Gauss)
saturation curve was measured

SUPS52m

Kharkov (Soviet) Supermendur
rolled and annealed at NSC KIPT

Weight: 4.383g
Length: 300.0mm
Width: 30mm
Thickness: 59.98 μ m(calculated)
Polarization: 7.953% (530Gauss)
saturation curve was measured

SUPA30m

American Supermendur
rolled and annealed at NSC KIPT

Weight: 2.644g(pol_calc)/2.633(label on foil)
Length: 299.0mm
Width: 30mm
Thickness: 36.30/36.15 μ m(calculated)
Polarization: 8.195% (530Gauss)
saturation curve was measured

H10

Kharkov (Soviet) Supermendur
rolled and annealed at NSC KIPT

Weight: 2.578g
Length: 36.2mm
Width: 30mm
Thickness: 32.87 μ m(calculated)
Polarization: 8.149% (530Gauss)
saturation curve was measured
was shipped to SLAC

H14

Kharkov (Soviet) Supermendur
rolled and annealed at NSC KIPT

Weight: 2.546g
Length: 36.2mm
Width: 30mm
Thickness: 32.46 μ m(calculated)
Polarization: 8.181% (530Gauss)
saturation curve was measured
was shipped to SLAC

IRON50m
American Fe rolled

Weight:	3.932g
Length:	300mm
Width:	30mm
Thickness:	46.26 μ m(calculated)
Polarization:	8.075% (530Gauss)
saturation curve was measured	